

Model validation

Modelling 2025

Class notes for 28 October 2025

[See also: Thackner et al. 2024.](#)

Concepts of Model Verification and Validation

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Plan for today

12.15-12.30 Questions on verification exercise

12.30-13.00 Validation demo

13.00-13.10 Break

13.10-14.00 Exercise

Validation is about assessing the accuracy of the model by comparison to the real world.

A "validated" model cannot be assumed to be accurate under all possible conditions, but it can show that the model is accurate enough for its intended use.

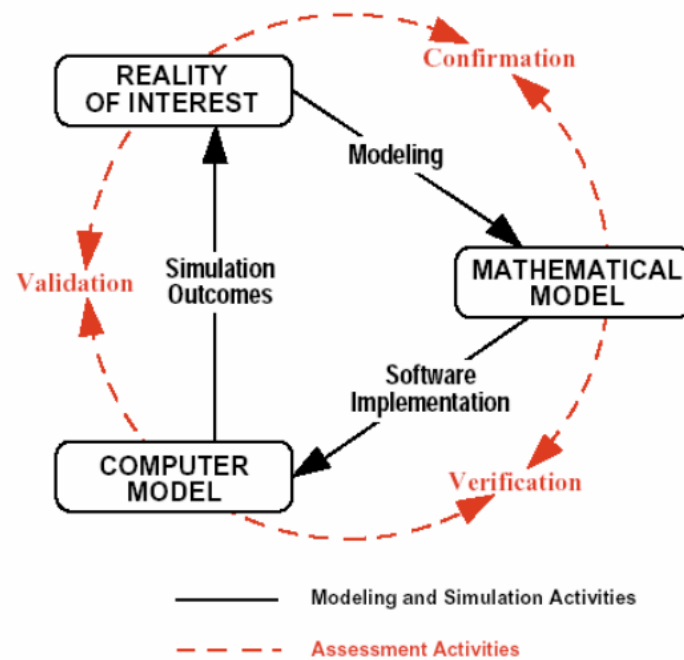


Figure 1: Simplified view of the model verification and validation process. [8]

From Thackner et al. (2024)

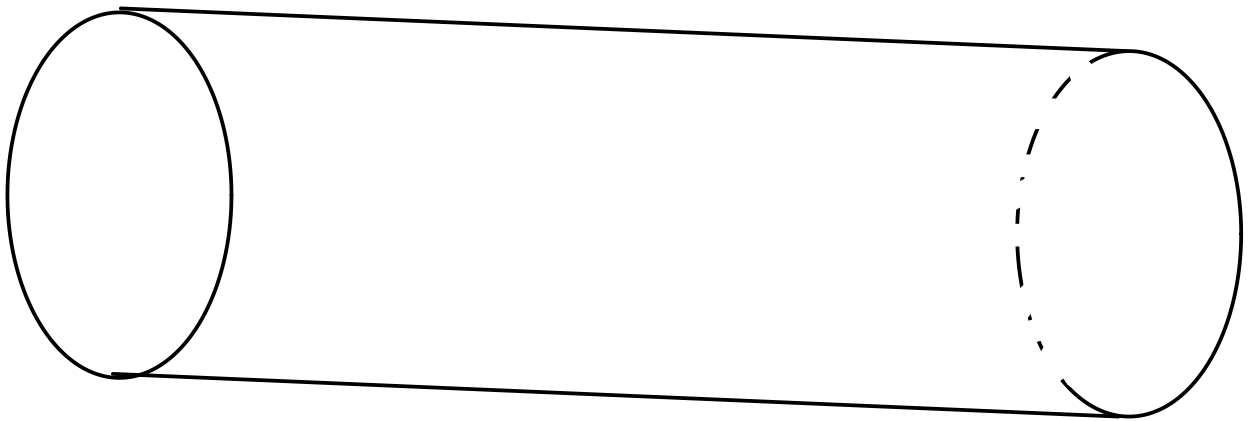
Model validation

Requirements:

- * Measurements and model outputs must be for the same thing (including time)
- * Application or conditions should be of interest (validation is specific)

Selection of a variable for validation depends on what is available and what is useful

Example: 1D diffusion



Comparing measurements and model output

We will focus on these approaches:

1. Graphical comparison: Plot both measurements and model output over time or space
2. Graphical evaluation of residuals: Look for patterns in difference between the two
3. Model fit or agreement statistics: mean absolute error (MAE) and mean bias error (MBE)

1 and 2 require alignment by time, so if measurements are not regularly spaced, the model function has to accept arbitrary times.

